Fiscal Year:	FY 2013	**Task Last Updated:**	FY 05/23/2013
PI Name:	Perera, Imara Y Ph.D.		
Project Title:	The Effect of Spaceflight on Transgenic Arabidopsis Plants With Compromised Signaling		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	27695	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	07/01/2010	**End Date:**	06/30/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-135 ; ISS NOTE: Extended to 6/30/2014 per A. Chu/ARC (Ed., 5/28/2013)		
Key Personnel Changes/Previous PI:	Caroline M Smith - Research Associate ; Eric S Land- Research Assistant		
COI Name (Institution):	Brown, Christopher (North Carolina State University) Sederoff, Heike (North Carolina State University)		
Grant/Contract No.:	NNX10AM72G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants provide a complete and economical means for human life support for long-term space exploration and habitation. However, since the space environment is not optimal for plant growth, an understanding of how plants sense and respond to changes in their environment is of fundamental importance. The phosphoinositide (PI) pathway is highly conserved among eukaryotes and functions in the regulation of a multitude of cellular pathways. The lipid-derived second messenger inositol 1,4,5-trisphosphate (InsP3) increases in response to many different stresses. We have shown that InsP3 levels increase with gravistimulation prior to visible bending in both monocot and dicot systems. We have generated transgenic Arabidopsis plants expressing the mammalian type I inositol polyphosphate 5-phosphatase (InsP 5-ptase), an enzyme that specifically hydrolyzes InsP3 and terminates the signal. The transgenic plants have normal growth and morphology; however, they exhibit altered responses to many environmental stimuli including gravity, drought and cold. While rapid changes in transcript levels occur in wild type Arabidopsis within 5 min of gravistimulation, the expression of several of the fastest responding genes does not change in the InsP 5-ptase roots in response to reorientation. Our hypothesis is that InsP3 is an important second messenger in the sensing and signaling of stimuli (including gravity). The plants with compromised InsP3 signaling therefore, provide a valuable tool for dissecting the role of the InsP3 pathway in plant responses to the microgravity environment encountered on the International Space Station. Our primary aim is to identify the molecular changes that are specifically mediated by InsP3 in the space environment. We will compare transcript and protein profiles of wild type and transgenic plants that are grown in both microgravity and 1g in space as well as on the ground. The long term goal is to understand the molecular mechanisms plants use to sense and respond to changes in their environment. This knowledge will help design plants which are better able to withstand space flight and microgravity conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The InsP 5-ptase transgenic plants provide a good model system for understanding several issues including plant responses to gravity, plant responses to environmental stress, and fundamental aspects of PI signaling. Our previous work has implicated InsP3 as a key player in gravity sensing and signaling and the transgenic plants have attenuated InsP3 signaling and reduced gravitropism. In addition these plants have altered responses to different environmental stresses. Therefore this study will help us understand plant signaling during environmental stress, which has long term implications with improving crops to better withstand unfavorable conditions. Furthermore, the phosphoinositide pathway is highly conserved among plants, yeast, and mammals. These studies using the transgenic plants with attenuated InsP3 signaling will provide valuable insights on the regulation and downstream effectors of this pathway and could also contribute to our understanding of the role of the PI pathway in regulating human physiology.
Task Progress:	1. The Plant signaling post-flight Ground Test was carried out at NASA ARC from 08/15/2012- 08/20/2012. The purpose of this test was to try and duplicate the environmental parameters encountered in the space flight (ramping up of CO2 levels). At the end of the test the samples were shipped to the PI lab for further analysis. 2. Work has been focused on growth measurements from the downloaded images obtained from the Plant signaling experiments. In particular, the goal is to compare root growth parameters between the 1g and micro g samples. Interestingly, preliminary results suggest that there were no obvious differences in root growth rates or cumulative growth between 1g and micro g samples; nor were there any significant differences between wild type and transgenic root growth (which is consistent with their growth on the ground). 3. The PI attended the ASGSR meeting in New Orleans, LA in November 2012; She served as a judge for the student poster competition. 4. The flight samples were finally transported back to Earth from the ISS on the Space X-2 on March 26, 2013. The samples were delivered safely to the PI's lab on April 2, 2013. In the PI's lab the seed cassettes were removed from the cold stowage bags, cataloged, and stored in the -80ºC freezer. Upon arrival in the PI's lab the samples were frozen and appeared to be intact. The major focus of the next few months will be to process the flight samples for RNA-Seq analysis to compare expression of wild type plants with the transgenic plants which have compromised InsP3 signaling.
Bibliography Type:	Description: (Last Updated: 05/30/2025)
Articles in Peer-reviewed Journals	Smith CM, Desai M, Land ES, Perera IY. "A role for lipid-mediated signaling in plant gravitropism." American Journal of Botany. 2013 Jan;100(1):153-60. http://dx.doi.org/10.3732/ajb.1200355 ; PubMed PMID: 23258369 , Jan-2013
Significant Media Coverage	Perera I. "A Lab Aloft (International Space Station Research): Growing Future Scientists with Plant Signaling Space Study. Article blog by Dr. Perera about her ISS Plant Signaling experiment and student involvement." http://blogs.nasa.gov/ISS_Science_Blog/2013/03/ , March 27, 2013., Mar-2013
Significant Media Coverage	UNC at work. @work in the world. "@work in the world: NC State Learning How to Grow Better Crops—From Outer Space. Article describing PI's flight experiment research on ISS." UNC at Work Newsletter. 2013 Apr;22., Apr-2013